

The Role of Technology and Innovation in Tackling Insecurity and Accelerating Development in Nigeria

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Abstract:

The perpetual rise in Insecurity and breakdown in the socio-economic development in Nigeria call for serious concern among researchers, policymakers, and the general public over decades. The wanton distractions of lives and properties posed by Boko Haram insurgency, banditry and kidnapping, pipeline vandalization, headsmen, and farmers clashes, as well as various crises in different parts of Nigeria, are spinning out of control which can lead to anarchy if not addressed and arrested. This paper, therefore, examines the role of technology and innovation in tackling insecurity and accelerating development in Nigeria. The study relied on secondary sources and data were sourced from journals, magazines, articles, textbooks, and personal experience. The Neo-Malthusian theory developed by Samuel Van Houten (1877) was used to analyze the reason behind ceaseless rises in insecurity and economic deterioration in Nigeria. As Nigeria is bedeviled with the challenges of insecurity, the paper found that Technology and Innovation would help Nigeria to counter and reduce its insecurity drastically, diversify its economy to mention just a few. However, the research identified some challenges that are militating against the innovation and technological advancement in Nigeria which include brain drain (migration of skilled personnel to developed countries), shortage of skills in science, research, and innovation, among others. To assure security and economic development in Nigeria therefore, the study recommends different regulations of combatting insecurity including human development-centered growth perspective and equitable allocation of resources, empower youths with the necessary skills, and paying more attention to innovations, technology, and implementation of existing research outcomes among others.

Keywords: Insecurity, Technology, Innovation, and development



INTRODUCTION

The growth and development of any society to a large extent depends on the degree of the security of lives and properties of its citizens and the ability to utilize its human and natural resources

effectively. The classification of Nigeria as an underdeveloped or third-world country because of its insecurity and economic deterioration despite its human and natural resources suggests

the fact that there are serious problems with the harnessing and utilization of the human and material resources in the country.

Since the return to democratic rule in May 1999, Nigeria has witnessed a multitude of security challenges related to militancy, kidnapping, armed robbery, political assassination, piracy, and ethnoreligious discords as well as arms proliferation. The challenges posed by insurgency in Nigeria have presumed a fearful extent forcing the country's political and economic officials and, indeed the whole nation, to rue the loss of their loved ones, investments, and lack of security in most of the country (Onifade, Imhonopi and Urim, 2013). The alarming level of insecurity in Nigeria has exacerbated the crime rate and attacks in many parts of the country, vacating unpalatable consequences for the nation's economy and its development. Different administrations since May 1999, have adopted several strategies aimed at alleviating the devastating consequence of insecurity without any positive result. When it appears that the government is winning in one side for instance Boko Haram, another will rare its ugly head such as banditry and kidnapping.

The motive behind government downfall to address this issue of insecurity satisfactorily may not be unconnected with government delinquency in dealing with the developmental desires of the people which has stricken them into establishing an alliance in different forms that have persisted in breeding what Onifade, Imhonopi, and Urim,(2013) refers to increasing ethnic hate, political rivalry, religious bigotry, and a rising population of disgruntled citizens in the country who feel they have been short-changed and given very restricted or no ticket to the common patrimony. This is compatible with the findings of Egwu, (2001) who observed that the primordial preferences of various ethnic factions towards chaos, the perennial outbreak of various ethnic militias, and the preponderant spiritual fundamentalism in place, given articulation to by some portions of the dominant religious organizations in Nigeria, have inescapably worsened the scale and propensity of insecurity and increased its spectrum in diverse ramifications.

Therefore, there is a need for the government and the general public to put all hands on the desk to minimize these challenges and accelerate Nigerian development. This is because one of the indices by which a nation's growth and development can be assessed is by its ability to protect the lives and properties of its citizens, its ability to harness its human and natural resources through innovation and technology to address its challenges, and accelerate its development. Thus, a nation's economic development is measured, determined, classified, compared, and ranked by its innovation and technological growth. In this sense, Nigeria as one of the most populous countries in the world with more than enough human and natural resources stands in an advantageous position to establish a strong digital economy sector, which will allow it to leapfrog its limitations and sustain its growth and development. Therefore, there is no doubt about the prospect of instantaneous technological advancement to help Nigeria confront the defining challenges of its time. So what is the role of technology and innovation in addressing insecurity challenges and accelerating development in Nigeria? To answer this question, this paper, therefore, examines the role of technology and innovation in tackling insecurity and accelerating development in Nigeria. However, before delving deep into the crux of the paper, essential knowledge of some crucial concepts is of paramount importance.

2:0. LITERATURE REVIEW

2:1. The concept of Insecurity

To understand clearly what insecurity entails it is imperative to know what is security. Security principally implies a sense of safety, protection from harm, or even survival in the face of some kind of menace. In the opinion of Nwanegbo and Odigbo, (2013), Nigeria's security involves efforts to strengthening the capacity of the Federal Republic of Nigeria so it can intensify its aim and objectives to thwart internal and external attack, eliminate corruption, restraint scandals, accelerate genuine development and progress as well as enhance the welfare and quality of life of every citizen.

On the other hand, insecurity according to Achumba, Ighomereho, and Akpor-Robaro (2013), involves susceptibility to harm, losing of life, property, or livelihood. Base on their perception insecurity is a situation of suspicion, a lack of control, and the incapability to take preventive efforts against adversaries that signify threat or devastation to an individual or group. In the context of this paper, insecurity is defined as an infringement of peace and security, whether religious, ethnonational, social, economic, and political that aids in recurring conflicts and leads to wanton devastation of lives and property. It is a state of uncertainty or anxiety due to the absence or depletion of protection. The issue of Insecurity has become a major challenge in Nigeria because of the various forms of carnage, brutality, pogroms, kidnapping, to mention just a few.

2:2. The concept of Technology

The word "Technology" was derived from two Greek words: "téchnē and logos" meaning the capacity of humans to create things using hands and/or engines (Terziovski, 2010). Technology can be defined as an invention using substances, tools, strategies, and sources of power to make life sustainable or more pleasant and work more productively. Technologies play a constantly intensifying role in all areas of social reality, which has jerked the foundations of social structures and systems and resulted in genuine changes in politics, economy, culture, and everyday life. It may also be defined as the effort to organize the world for problem-solving so that goods and services can be developed, produced, and used (Lemelson-MIT Program, 2004).

2:3. The concept of Innovation

Innovation as a concept is difficult to define, but economists generally use the term to refer to increases in the quality and variety, or reductions in the cost, of goods and services provided by the market (Broughel & Thierer, 2019). Innovation is about excavating new and better ways of living and bringing up new ideas or new types of products and services into existence, it can also be changes in manufacturing strategies, minor improvements, adjustments, and refinements to existing products. It involved a complex process of introducing new ideas into use. Sofia Ranchordás (2015), argued that "Innovation is more than the latest technology, it is a phenomenon that can result in the improvement of living conditions of people and strengthening of communities.

2:4. The concept of development

Despite the intensifying interest in the concept of development from the academic sphere, the concept is notoriously difficult to define. The meaning and the conceptualization of the term were greatly influenced by the ideological contradiction between the

Capitalist West and the Socialist East. While some people see development as Industrialisation and Modernity or even Westernisation and its artifacts such as cars, trains, phone-sets, television, computers, schools, radio, electricity, textiles, tarred roads, etc. To Rodney (1972), development is the improvement and progress of the individual, skills, and capacity, greater freedom, creativity, self-discipline, responsibility, and wellbeing of the people. He also sees development as an overall social procedure which is dependent upon the high capacity of members of a society to master the laws of nature (that is science) and assign such laws in the generation of tools (that is technology) with which they can manipulate their environment to meet their immediate and future needs. Thus, according to Rodney development entails both quantitative and qualitative improvement in all spheres of a given society and it cannot be seen purely as an economic affair because other segments of the society are also involved.

Morris (2010), perceived development as the bundle of technological, subsistence, organizational, and cultural accomplishment through which people house, clothe, feed, and reproduce themselves, resolve disputes within their communities, explain the world around them, and extend their powers at the expense of other communities as well as defend themselves against others' attempt to extend power. Therefore, development can be defined as advancement through progressive changes in the economic, technological, cultural, and political conditions of a society leading to an improvement in the welfare of the citizens.

3.1. THEORETICAL FRAMEWORK

This study is anchored on Neo-Malthusian theory which has the same view as the Malthusian theory. The theory concentrates on the interconnectedness between growth of population and food mandated for subsistence. The advocates among the Neo-Malthusians are Samuel Van Houten, Paul Ehrlich, and Garrett Hardin.

Thomas Malthus argued that populations grew in a geometric sequence which means the progression of numbers where each term after the first is found by subtracting the previous one by a fixed, non-zero number called the common ratio. For instance, in the sequence 2, 10, 50, 250, the common ratio is 5. Furthermore, he stated that food production improves in arithmetic progression which means a sequence of numbers in which the disparity between the consecutive terms is constant. For example, in sequels 2, 5, 8, 11, 14, the common distinction is 3. Base on this analysis we can conclude that populations will grow faster than the supply of food. This consequently will lead to a deficit of food which eventually leads to conflict over the sacred resources.

Ehrlich (1968) recapitulates this position by explaining that; too many people couple with too little food resulted in conflict over sacred food and lead to deterioration of environment. Homer-Dixon (1999) expanded the view of Ehrlich (1968) that as a consequence of the scarcity of resources constantly fuel cut-throat conflict among different actors, which ultimately leads to social conflict that endangers national security due to escalation of revolts from actors, interest groups, and nations. The relevance of this theory as detected by Adebakin and Raimi, (2012) is that the increasing population melted with scarce natural resources jeopardizes national security as a result of cut-

throat competition between and among the interest groups in society. This various social group uses both legitimate and illegitimate means to obtain power over insufficient resources. However, the critics of the theory is that the prediction put forward by Malthus for the geometric growth of populations and the arithmetic growth of food have not played out. In Western Europe for instance, populations have grown contrary to Malthus prediction and food production has also increased because of advancements in technology and innovation.

The applicability of this theory to this study is fact that the theory succinctly explains the factors responsible for the magnitude of insecurity in Nigeria by summed up the consequences of the persisting population growth and the agitation for resource control in Niger-Delta and destructive agitation by Boko Haram for the establishment of shari'ah states in the Northern Nigeria to control the resources.

4.1. Causes and consequences of insecurity in Nigeria

Many scholars have pointed out several causes of conflict and insecurity in Nigeria that are inimical to socio-economic and national development. These causes have been categorized into external and internal factors. In this paper, we are going to focus on internal causes because the internal causes of insecurity in Nigeria cause a major challenge to socio-economic and political development than the outer causes of insecurity. These causes include:

Moral and values depletion, as well as insufficient and wrong religious education, are the major causes of unrest and the generally miserable state of security of Nigeria as a nation-state. It is unimaginable to expect anything good to come from an individual that is morally bankrupt and has wrong or negative values. An orange tree cannot produce an apple, it is practically impossible. The behavioral offspring of such morally deficient individuals is the aspiration for a luxurious life without passing through hardship. Therefore, they can do whatever their bad attitudes ask them to do which consequently contributed immensely to the worst security situation that Nigeria found itself in.

Secondly, youths' restiveness has been another cause of insecurity in Nigeria. The high rate of unemployment, poverty, and economic deprivation have forced many youths to engage in criminal activities. Different surveys' conducted by various scholars disclosed that economic deprivation and poverty are the major causes responsible for Nigerian insecurity, this is because terrorists are products of unemployment and are easily recruited by sponsors. It has been reported by the National Bureau of Statistics that Nigeria's unemployment rate in August 2020 is 27.1% which means about 21.7 million Nigerians remain unemployed. It is of serious concern a country that has over 250 mineral resources government fails in providing job opportunities for its populace. On the other hand, although there is a lack of sport from the government, the youths also have their lopsidedness for being so lazy and utopian aspirations of having luxurious life and looking for government to provide everything for them, this has eventually brought Nigerian development into a halt.

Thirdly, irresponsible leadership is amongst the causes of the Nigerian tragedy. The politics of prebendalism and lack of savvy in politics have led Nigeria to become a playground of terrorists, this happened as a result of mismanagement and pervasive

corruption that spread to all nooks and cranny of the country. The reason is that the country's resources that suppose to be used in creating opportunities and empowering youths have been hijacked and orchestrated by the cabals and political prebendalists. Other factors responsible for Nigerian insecurity include extraneous ideology and ignorance, weakness of the government to provide effective security as well as her inability to follow due process in identifying and punishing culprits, feeling of hopelessness, military/police high handedness, hatred of the government, lack of basic amenities and security lapses among others.

The unpalatable consequences of insecurity in Nigeria and its prevalent cost are difficult to measure in monetary or real terms, however, it is obvious to observers that it has cost Nigerians enormous human and material resources. It is crystal clear that insecurity is ceaselessly retarding the economic and technological advancement of Nigeria. Many Banks and places have been attacked by terrorists which destroyed many business centers and necessitated foreign investors to be scared away from Nigeria. Apart from destructions of properties, business centers, and fear of investments by entrepreneurs, if you calculate the budgetary allocation to security sector, from 2010 to 2020 Nigerian government have allocated over N8 trillions to the security sector. Imagine if these trillions of naira have been used for youths empowerment and nation-building how far Nigeria will reach in terms of development? Besides, many productive lives that could be used for nation-building have been lost, it is estimated that terror groups have killed over 30,000 people in Nigeria since 2003, causing the displacement of more than 2.4 million people. This means that if it persists, Nigeria has no future because her intellectuals are constantly being killed, the available ones are displaced and live in constant fears, uncertainty, and pain of the loss of loved ones. If this continues then what is the future of Nigeria as a country?

5.1. THE ROLES OF TECHNOLOGY AND INNOVATION IN TACKLING INSECURITY AND ACCELERATING DEVELOPMENT IN NIGERIA.

Evidence in the economic literature indicates that technology and innovation are the cornerstones of sustained economic growth and prosperity. With Nigerian powerful entrepreneurial climate, the narrative that technological innovation and digital disruption can minimize Nigerian insecurity and spark its economic growth is persuasive—especially through the success stories of using sophisticated weapons and software for mass surveillance, mobile money, as well as agricultural extension and health services supported by technology and innovation.

To be convinced by this assertion, this paper argues that technology and innovation can help Nigeria to counter and reduce its insecurity drastically by using sophisticated facial recognition software for mass surveillance. They would help Nigeria in surveilling and eliminating terrorists particularly that of Boko Haram in the Northern part of the country. Using Information technology (IT) for instance will play a vital role in strengthening Nigeria's National security against potential future attacks and threats by providing authentication, containment, privacy, detection and identification, image and video processing, and evidence combination; identify potential threats, share information more readily, provide mechanisms to protect

the Nation, and develop response capabilities. Also, part of the reason why Nigeria is unable to deal with the terrorists decisively is because of its inadequate military weapons to be used by security forces. By creating Nigerian-based weapons, security agencies will be more equipped with devastating weapons which will enable them to reduce the terrorists drastically.

Secondly, Technology and Innovation would contribute immensely to the diversification of the Nigerian economy which in turn will play an important role in increasing wealth and development. A nation that has different sources through which it earns its income has enormous potentials to win against the risk of failure in its economy. Nigeria with more than enough human and natural resources has the potential to diversify its economy. The various types of natural resources could be used in inventing different types of products. For example, instead of Nigeria exporting petroleum resources, it will produce jewelry products and export them as finished goods. By well harnessing technological advances and constant innovations, many opportunities would be unveiled capable of diversifying the Nigerian economy which will lead to alleviating, if not eradicating the high level of poverty and insecurity; alongside increasing access to quality health care, education, and infrastructure in the country. This, in turn, will provide a conducive environment in which innovative solutions to local challenges can be showcased.

Thirdly, with the high rate of over 200 million people in Nigeria, there is a need for enough food that will feed the country to live in peace. The Nigerian agricultural climate is highly fertile, and it is rich in biodiversity, the agricultural environment across the country can cultivate and support almost all the types of crops that are being cultivated in the world. Looking at this favorable agricultural atmosphere, Nigeria stands in an advantageous position to feed its population and generate income through agricultural production. However, despite this natural endowment the country's food production is declining. Also, her status as an agricultural powerhouse in the world has deteriorated. Again, while Nigeria is regarded as the largest producer of cassava in the past, now the outcome per hectare comes to be one of the lowest in the world. However, principally due to lack of innovation and poor technological development, today Nigeria has deviated from being a self-sufficient country in food to being a food importer. Therefore, assigning a wide range of technical innovations such as irrigation, plough, mills, crop rotations, chemical fertilizers, and pesticides, and many more will lead to increased productivity of the production factors of land, labor, and capital in Nigeria. For example, in the 1930s in the US, 25% of their people worked in the agricultural sector while the entire Gross Domestic Product (GDP) was below \$100 billion. Today, less than 2% of the US population works in the agricultural sector, while the whole GDP is over \$14 trillion. Thus, technology and innovation will drastically increase the agricultural output of Nigeria which eventually will provide enough food for Nigerians and also export its for income.

Lastly, the role of Technology and Innovation cannot be overemphasized in terms of creating job opportunities, providing a standard of living, and poverty reduction. Innovation in the 21st century is what produces wealth and creates jobs. History is quite clear on the significance of innovation to sustained job creation. Those nations that have leveraged the

fruits of science and technology for greater socio-economic development have led the world in well-paying jobs and a standard of living for their citizens. Part of the challenges facing Nigeria is its inability to provide job opportunities for its citizens which arguably resulted in insecurity. Therefore, Nigeria should concentrate more on innovation and technology, this of course will lead to industrialization which people would get a chance to be employed, and the insecurity challenge particularly in the northern part of the country where they lack industries will be reduced to the lowest ebb.

However, it has to be noted that many factors are militating against Nigeria's innovation and technological advancement. It takes time to learn how to assign new technologies effectively and that discovering takes resources and a huge amount of human capital. The challenges include brain drain (flight of skilled personnel to other parts of the world), insufficient levels of literacy, inadequate infrastructure, and a shortage of skills in science, research, and innovation, inadequate funding, etc. Until and unless the above challenges among others have been addressed, the aforementioned roles can not be well harnessed.

6:1. CONCLUSION

Security, without doubt, is the sine qua non for the credible existence of human beings, a society, its harmony, and economic advancement as well as political stability. Looking at the aforesaid, Nigeria's greatest hope for addressing its challenges particularly that of Boko Haram, and accelerating its growth and development lies beneath its vibrant human and natural resources. To achieve this aim, these resources must be empowered with the necessary skills and more attention must be paid to innovations, technology, and implementation of existing research outcomes. It is crystal clear that technology and Innovation played a significant role in many countries in the world like the US, Britain, France, China, etc, and make a lot of contributions in providing employment opportunities, counter-terrorism, business expansion, and agricultural modernization, among others. That is why growth economists, development economists, and economic historians all appear to agree that technology and innovation are the major determinants of societal growth. Therefore Technology and Innovation can tackle Nigeria's insecurity and accelerate its growth and development if we apply them decisively.

7:1. RECOMMENDATIONS

From the foregoing discussion, the following recommendations are provided:

- i. The government at all levels must be proactive in dealing with security issues and threats, through deploying advanced technology, modern and logistics methods of training, and motivation of the security personnel;
- ii. For security challenges to become history in Nigeria, the government must tackle the economic bottleneck and accelerate the pace of development. The pace of development include social, economic, political, and physical infrastructure for business activities and industrial growth, to provide gainful employment, medical care for the people, and a high level of educational facilities;
- iii. Nigeria should unite and make an ethical revolution on over-reliance on natural resources and passive reliance on

international aid and build innovative minds; and decisively channel national capacities and energy into building a strong, sustainable, innovation-based, Nigeria-centric economy for current and future generations;

- iv. Nigeria's innovation policy should enhance returns to investment in R&D, and center on many critical policy domains including economic, regional, and industry development, science, and technology policy, labor market and industrial relations policy, education policy, etc.
- v. Nigeria should establish Technology Innovation Support Centers and Technology Transfer Offices in universities and R&D centers as well as mentorship programs to identify potential young innovators and build their capacities.

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